# Project Report: Brute-Forcing FTP and SSH Services Using Hydra

**1.Live Host Discovery**

Using the sudo ifconfig command, I assigned IP addresses to the Kali (192.168.1.4) and Alice (192.168.1.2) systems, ensuring they were on the same network. I verified connectivity by successfully pinging Alice from Kali. Next, I performed a network scan using **Nmap** to identify live hosts on the network. This scan revealed the presence of Alice at IP address **192.168.1.2**. An Nmap scan on Alice identified two open ports:

Port 21( **ftp**)

Port 22 (ssh)

These open ports indicated the presence of potential services for further investigation.

**2.Brute-Forcing Credentials with Hydra**

I created two files, one containing potential usernames and another with passwords, using the **touch** and **nano** commands. The contents of these files were verified using the **cat** command. Below are the files created:

Usernames file (**username**)

Passwords file (**password)**

Hydra was then used to perform a brute-force attack on the FTP service. The following command was executed**:**

**hydra -L username -P password [ftp://alice](ftp://alice)**

**Hydra successfully cracked the credentials,Username: alice ,Password: alice**

## For SSH Service

Attempting to brute-force the SSH service using Hydra resulted in an error, likely caused by firewall restrictions or security configurations on the target system.

**3.Analysis & Recommendations**

This project demonstrated how weak or default credentials can be exploited to gain unauthorized access to network services. Based on the findings, the following security measures are recommended to harden the system and prevent brute-force attacks:

1. **Use Strong Passwords:** Ensure all accounts use complex passwords with a mix of letters, numbers, and symbols.
2. **Limit Login Attempts:** Configure services to lock accounts or block IP addresses after a certain number of failed login attempts.
3. **Implement Multi-Factor Authentication (MFA):** Add an additional layer of security by requiring a second form of authentication.
4. **Regular Updates:** Keep software and services updated to patch known vulnerabilities.